Data Science & ML Quick Reference Manual

NumPy Array Vectorization, Pandas DataFrames & Model Evaluation

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1. NumPy Vectorization & Array Operations

- Always avoid raw Python for-loops over numeric matrices.
- Broadcasting: automatically expand arrays with compatible dimensions.
- Boolean indexing: df[df['score'] > 85] for blazingly fast filtering.

2. Exploratory Data Analysis with Pandas

- df.info() and df.describe() for quick statistical distributions.
- Handling nulls: df.dropna() vs df.fillna(df.mean()).
- Groupby operations: df.groupby('department')['rating'].agg(['mean', 'count']).

3. Supervised Model Evaluation

- Classification metrics: Accuracy, Precision, Recall, F1-Score, ROC-AUC.
- Cross-validation: 5-fold KFold split to prevent overfitting.